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DATA DESCRIPTOR

Comprehensive Dataset for Event Classification Using Distributed Acoustic Sensing (DAS) Systems

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Distributed Acoustic Sensing (DAS) technology leverages optical fibers to detect acoustic signals over long distances, offering high-resolution data critical for applications such as seismic monitoring, structural health monitoring, and security. A significant challenge in DAS systems is the accurate classification of detected events, which is crucial for their reliability. Traditional signal processing methods often struggle with the high-dimensional, noisy data produced by DAS systems, making advanced machine learning techniques essential for improved event classification. However, the lack of large, high-quality datasets has hindered progress. In this study, we present a comprehensive labeled dataset of DAS measurements collected around a university campus, featuring events such as walking, running, and vehicular movement, as well as potential security threats. This dataset provides a valuable resource for developing and validating machine learning models, enabling more accurate and automated event classification. The quality of the dataset is demonstrated through the successful training of a Convolutional Neural Network (CNN).

Background & Summary

Distributed Acoustic Sensing (DAS) is an emerging technology that utilizes optical fibers as sensing elements to capture acoustic signals along the length of the fiber. This technology has gained significant attention because it provides high-resolution spatial and temporal data across extensive distances. DAS has found applications in various fields, including seismic monitoring^{1,2}, structural health monitoring³, pipeline surveillance⁴, and railway safety⁵. However, one of the key challenges in deploying DAS systems for practical applications is the effective classification of events detected by the sensor. Accurately distinguishing between different types of events, such as seismic activities, vehicular movements, and human activities, is crucial for the technology's reliability and broader adoption⁶.

This task of event classification is particularly challenging due to the complexity and variability of signals captured by DAS systems. These systems record a vast array of signals that can originate from diverse sources, including environmental noise, human activities, and mechanical vibrations. The difficulty lies in accurately distinguishing between these different types of events, especially when they exhibit overlapping or weak signal patterns. Traditional signal processing methods often struggle to handle the high-dimensional, noisy data generated by DAS systems.

The high dimensionality arises from the large number of sensing points along the fiber, resulting in voluminous and complex datasets. Conventional signal processing techniques, typically designed for lower-dimensional data, struggle to efficiently manage and extract meaningful patterns from such massive data streams⁷. The computational complexity and memory requirements associated with processing DAS data further exacerbate these challenges, limiting the scalability of traditional approaches.

In addition to high dimensionality, DAS data is prone to various noise sources, including environmental noise, mechanical vibrations, and inherent instrument noise from DAS interrogators. This contamination can significantly degrade the quality of acquired signals, making it difficult to distinguish between true events and background noise. Traditional denoising methods, such as wavelet-based filtering or adaptive thresholding, often prove inadequate in addressing the complex and highly variable noise characteristics of DAS signals⁸.

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Consequently, there is a growing need for advanced techniques that can enhance signal quality while preserving critical event information.

Φ -OTDR systems rely on highly coherent laser sources to maintain phase stability and minimize signal drift. However, phase noise, including modulation instability (MI), can degrade performance. Techniques like coherent seed injection help suppress MI-induced noise, enhancing robustness against environmental fluctuations⁹. For anti-interference, Φ -OTDR benefits from distributed sensing, allowing spatially resolved event detection. Deep learning-based denoising improves signal clarity by enhancing the signal-to-noise ratio and reducing false positives¹⁰.

To overcome these challenges, machine learning techniques have become increasingly important⁷. Capable of learning complex patterns and correlations within the data, these methods enable more accurate and automated classification of events. By leveraging large datasets and advanced algorithms, machine learning approaches significantly enhance the reliability of DAS-based event detection and classification, making them a crucial tool for unlocking the full potential of DAS technology in real-world applications^{8,11}.

A major challenge in advancing event classification for DAS systems is the need for large, high-quality datasets¹². Obtaining such datasets can be difficult due to several factors. DAS technology is very expensive, and access to data is often limited to organizations with the resources to deploy and maintain these systems. Additionally, some DAS measurements may contain sensitive information, such as data from ocean-based systems that could inadvertently capture the locations of submarines or other classified activities. As a result, there is a significant lack of publicly available datasets, which has led researchers to rely on artificially generated data to train and validate their models. Recognizing this gap, we have compiled and provided a labeled dataset of DAS measurements collected around a university campus. This dataset offers a valuable resource for the research community, enabling the development and testing of more accurate event classification algorithms in a controlled, yet realistic, environment. The quality of this dataset has been validated through the successful application of a Convolutional Neural Network (CNN) model. This high level of accuracy underscores the dataset's robustness and its suitability for training machine learning models, further contributing to the advancement of DAS technology¹³.

Methods

Phase-Sensitive Optical Time-Domain Reflectometry (Φ -OTDR) is a specific type of DAS system that leverages phase-sensitive technology to enhance the detection and monitoring capabilities of standard DAS systems. Φ -OTDR provides high sensitivity and spatial resolution, making it ideal for precise and detailed distributed sensing applications¹⁴.

Φ -OTDR operates by sending highly coherent laser pulses into an optical fiber. As these pulses travel through the fiber, they are scattered back by Rayleigh scattering, which are due to small imperfections within the fiber¹⁵. The coherence of the laser is crucial because it allows the system to detect phase changes in the backscattered light, which are induced by dynamic strains or vibrations along the fiber.

The coherent light pulses cause interference effects in the backscattered signal, making the system extremely sensitive to phase changes. When a dynamic strain or vibration occurs along the fiber, it alters the phase of the backscattered light. By measuring these phase changes, Φ -OTDR can detect very small disturbances with high spatial resolution. The resulting interference patterns are analyzed to determine the location and magnitude of the disturbances, providing detailed information about acoustic or vibrational events along the fiber.

Φ -OTDR is used in a wide range of applications due to its high sensitivity and resolution. In seismic monitoring, it can detect minute ground movements and provide valuable data for earthquake detection and geophysical exploration¹⁶. In pipeline monitoring, Φ -OTDR can identify leaks and unauthorized activities with great precision^{17,18}. The technology is also employed for vibration analysis in industrial equipment and for detailed structural health monitoring of buildings and infrastructure.

One of the significant advantages of Φ -OTDR as a DAS system is its ability to monitor long distances with a single fiber, making it cost-effective for extensive monitoring applications¹⁹. It provides continuous, real-time monitoring, essential for high-resolution and high-sensitivity detection applications.

The dataset was collected using an Optasense ODH-F DAS interrogator (more details available at <https://www.optasense.com/technology/odhf/>), which operates based on Φ -OTDR. This system analyzes the most common standard ITU-T G.652.D single-mode fiber link, initially intended for inter-campus communication. The fiber is installed 1 meter underground, adjacent to the pavement, as illustrated in Fig. 1, creating an ideal setup for distributed sensor experiments.

Data acquisition was conducted primarily under sunny weather conditions in April 2023, ensuring minimal noise interference from environmental factors, such as rain. This allowed for the collection of clean acoustic data, reducing the likelihood of weather-related anomalies in the dataset. The ambient temperature during data collection ranged from 18 to 22 °C, reflecting typical campus weather during the recording period. The temperature variations within this range were slow and gradual. They did not introduce any detectable signatures or extra noise in the measurements, further contributing to the overall quality and reliability of the dataset.

The DAS system is configured with a pulse repetition rate of 20 kHz, allowing it to effectively capture rapid changes in the acoustic environment. The system utilizes laser pulses with a width of 20 ns, providing high-resolution spatial sampling. The spatial sampling interval is set to 1 meter, enabling precise localization of events along the fiber. The system operates at a wavelength of 1550 nm, which is optimal for minimizing attenuation and maximizing the signal-to-noise ratio in standard single-mode fibers. The maximum number of channels configured in the system is 1663, allowing for only local coverage along the fiber's length.

To ensure comprehensive data capture, a DAS operator was stationed in a campus building with a view of the analyzed street. The operator utilized Optasense software to manage the data acquisition process. During the data collection, several events of interest, including walking, running, and car driving, were deliberately



Fig. 1 Aerial view of the university campus showing the layout of the distributed acoustic sensing (DAS) system. The green line indicates the location of the sensing fiber, which is buried 1 meter underground alongside the pavement. Various events occurring near the fiber are highlighted, including regular activities such as walking, running, longboarding, and driving, as well as potential security-related events like fence climbing, fiber manipulation, and opening/closing of manholes. The DAS interrogator is positioned as shown, monitoring these events for analysis.

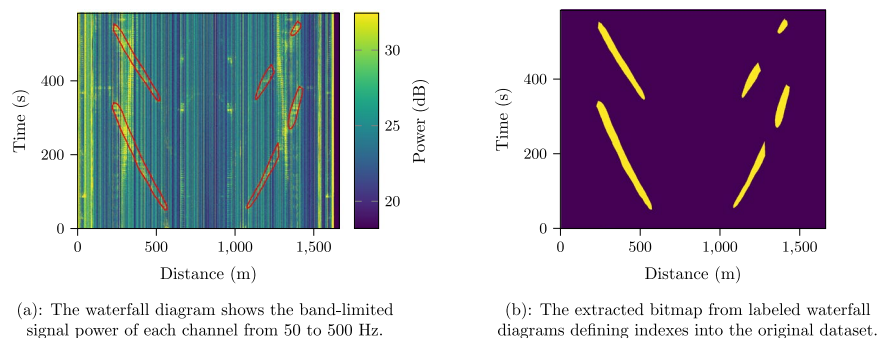


Fig. 2 The captured datasets are analyzed by evaluating the average signal power of a certain frequency range used for manual dataset labeling.

generated by colleagues to provide a diverse range of vibration signatures as labeled data. These controlled activities were conducted at various times and locations along the fiber's length to simulate real-world scenarios and assess the system's performance in detecting and analyzing different types of disturbances.

Data Records

The dataset is publicly available on Figshare²⁰. The dataset predominantly includes events relevant to its location, capturing various activities commonly performed by students. These regular events encompass driving cars, running, walking on campus, and commuting using longboards. In addition to these routine activities, the dataset also contains events that could indicate potential security threats. These include actions such as climbing and knocking on fences, tampering with fiber cables, and opening and closing manholes where the fibers are housed. Furthermore, the dataset includes records of construction activities, specifically the construction work for a new laboratory in a campus building. All dataset events have specific labels highlighted by a yellow color in Fig. 1.

Labeling. Each measured event is represented by one or more DAS data files in the **HDF5** format, identified by the *.h5 suffix. Each file contains N channels with 1D acoustic signals. However, raw acoustic channels are challenging to visualize directly. Therefore, the data are represented using band-limited signal power P_{band} , effectively highlighting events occurring near the fiber, as illustrated in Fig. 2a.

The band-limited signal power P_{band} (in dB) is computed as

Label	Event count	HDF5 file count	Size (GB)
Car	20957	3	12.6
Construction	29169	1	2.4
Fence	3601	3	3.7
Longboard	11908	1	5.5
Manipulation	17223	1	3.7
Open/close manhole	5943	1	1.9
Regular	210023	1	5.7
Running	12927	1	2.7
Walk	12714	2	8.8

Table 1. Number of events for each label in the dataset, along with the corresponding number of files and total data size for each label.

$$P_{\text{band}} = 10 \log_{10} \sum_{k=k_{\min}}^{k_{\max}} S_x(f_k), \quad (1)$$

where $S_x(f_k)$ is a one-sided estimate of power spectral density computed using the squared magnitudes of the Discrete Fourier Transform (DFT) coefficients and $k_{\min}$ and $k_{\max}$ are indices such that $k_{\min} = \min\{k|f_k \geq f_{\min}\}$ and $k_{\max} = \max\{k|f_k \leq f_{\max}\}$, respectively.

The labeling process involves utilizing these P_{band} figures to identify and highlight relevant data vectors. To ensure consistency and prevent variability between different annotators, a single expert performed the entire annotation process. Borders were drawn on a graphical tablet, marked with red lines, to indicate usable data vectors. This process resulted in a bitmap file with a `*.npy` suffix, which maintains the exact dimensions as the original P_{band} figure. This bitmap file, shown in Fig. 2b, is used to differentiate between irrelevant vectors and those containing the acoustic signatures of the analyzed event.

While manual annotation inherently involves some subjectivity, the use of a single annotator eliminates inter-annotator variability, ensuring a uniform standard across the dataset. Additionally, to further refine the labeled dataset and minimize potential boundary inconsistencies, a post-processing step was applied. The lines drawn in the P_{band} figures may not perfectly align with the borders of occurring events. Therefore, some of these vectors were filtered out using a spectral condition. This process involved converting a data window into a spectrum using a fast Fourier transform (FFT). The condition then compared the difference between the lower and upper parts of the spectrum, allowing for the exclusion of irrelevant vectors that contained a significant amount of noise, where their spectrum was nearly flat. The resulting number of samples per label, together with the size of raw HDF5 files, is shown in Table 1.

The unlabeled data can be accessed directly via HDF5 files, which are organized into subdirectories named after their respective labels. These files contain a 1D acoustic signal for N channels, as described in earlier sections. The 1D signal can be further divided into smaller segments using a sliding window approach, similar to the labeled data option.

The labeled dataset files additionally include bitmap data stored in `numpy` files. During the labeling process, the 1D signal from each channel was segmented into blocks of 8192 samples, using a sliding window with a 2048-sample shift. When working with the labeled dataset, it is crucial to use the same window size of 8192 and window shift of 2048 to ensure that the bitmap aligns correctly with the data.

Dataset Folder Structure. The dataset²⁰ is organized into a structured directory format to facilitate reproducibility and ease of use. The top-level folder, `DAS-dataset`, contains all data files and supporting scripts. Within the dataset, the `data` subdirectory houses categorized event recordings, with separate folders for different types of activities such as walking, running, and vehicular movement. A `python` directory is also provided to support data loading and preprocessing.

Three types of files represent each recorded event: **HDF5 files** (`.h5`) - These contain the raw data collected from the OptaSense system, including distributed acoustic signals across multiple sensing channels. **JSON files** (`.json`) - These store annotation metadata, specifically information about manually drawn curves during the labeling process, which highlight the event's occurrence. **NumPy files** (`.npy`) - These contain bitmap representations that specify which signal windows contain or do not contain an event. This binary mask is used for event localization and training machine learning models.

The dataset hierarchy is outlined below:

- `DAS-dataset/` (Root directory)
 - `data/` (Contains the main dataset files)
 - `car/` (Events related to vehicles)

- `auto_YYYY-MM-DDThhmmss.h5` - **Raw DAS data** from the OptaSense system.
- `auto_YYYY-MM-DDThhmmss.json` - **Annotation data** with manually drawn labeling curves.
- `auto_YYYY-MM-DDThhmmss.npy` - **Bitmap representation** indicating event presence per window.
- `construction/` (Construction activity events)
- `fence/` (Security-related fence events)
- `longboard/` (Longboarding activities)
- `manipulation/` (Fiber cable manipulation events)
- `openclose/` (Opening/closing manhole events)
- `regular/` (General background activity)
- `running/` (Human running activities)
- `walk/` (Walking activity events)
- `python/` (Scripts for dataset handling)
 - `example.py` (Sample code for dataset loading)
 - `Pipfile` (Dependencies for Python environment)
 - `Pipfile.lock` (Locked dependencies)
 - `LICENSE` (Licensing information)

HDF5 Data Structure. The raw DAS data collected in this study is stored in HDF5 format (`.h5`), which provides a hierarchical structure for efficient storage and retrieval. This format enables the structured representation of raw sensor signals and metadata attributes. Below, we describe the key elements of the dataset structure.

Each HDF5 file follows a hierarchical grouping system that categorizes the data into logical sections. The main groups and their respective subgroups are as follows:

- `/` (Root group)
- `/Acquisition` (Main acquisition metadata)
 - `/Acquisition/Custom` (Custom acquisition attributes)
 - `/Acquisition/Raw[0]` (Primary raw data storage)
 - `/Acquisition/Raw[0]/Custom` (Custom metadata for raw data)
 - `/Acquisition/Raw[0]/RawData` (Recorded signal data)
 - `/Acquisition/Raw[0]/RawDataTime` (Timestamps for raw data)

Each of these groups contains datasets and attributes relevant to the DAS system's operation. Table 2 summarizes the key datasets and attributes stored in the HDF5 files, including acquisition settings, raw data format, and metadata necessary for data interpretation.

Technical Validation

The technical validation of the dataset was primarily established by successfully training a simple Convolutional Neural Network (CNN) model, which consisted of two convolutional layers with 64 and 256 filters and two dense layers with 1024 and 7 neurons, respectively. Using this dataset, the model achieved an accuracy of up to 91 %. This high accuracy indicates that the dataset is of sufficient quality and robustness for training machine learning models, specifically in the context of DAS data analysis. For more details about the methodology and the CNN model itself, we refer the reader to paper¹³.

The dataset includes various labeled events, and the CNN's ability to accurately classify these events demonstrates that the data is both relevant and representative of the activities it is intended to capture. The high accuracy also suggests that the dataset contains clear and distinguishable features for each class, which is critical for the effective training of machine learning models.

To prepare the data for CNN training, each channel of the raw DAS signals was first windowed using an 8192-samples-long Hann window and transformed into a frequency-domain representation using an oversampled Discrete Fourier Transform (DFT) with an oversampling factor of 3. This resulted in a three-dimensional magnitude spectrum ($channel \times time - window \times frequency$), capturing key frequency components relevant for classification. To reduce redundancy, only the first 2048 elements of the single-sided spectrum were retained, as lower frequencies contained the most informative signal characteristics. These processed spectral representations served as input to the CNN, providing a structured and informative feature space that enabled the model to learn distinct event patterns effectively. Additionally, a band-limited signal power P_{band} between 50 and 500 Hz was computed, generating a two-dimensional matrix used exclusively for manual labeling. These manually annotated bitmaps then served as labels for model training.

During the model training process, the dataset was split into training, validation, and testing in an 80:10:10 ratio to assess the model's generalization capability. As shown in Table 3, the test metrics confirm that the data is not only well-labeled but also free from significant noise or inconsistencies that could hinder model performance. The high accuracy indicates reliable classification, while the strong precision and recall scores demonstrate the model's balanced performance in minimizing both false positives and false negatives. Overall, the successful training of a CNN with high accuracy is strong evidence of the dataset's technical validity. It

Group / Dataset	Description	Data Type / Shape
/	Root group containing metadata	—
/Acquisition	Acquisition metadata	—
/Acquisition/AcquisitionId	Unique identifier for acquisition	String
/Acquisition/MaximumFrequency	Maximum sampled frequency (Hz)	Float (10,000)
/Acquisition/MinimumFrequency	Minimum sampled frequency (Hz)	Float (0)
/Acquisition/PulseRate	Pulse repetition rate (Hz)	Float (20,000)
/Acquisition/PulseWidth	Laser pulse width (ns)	Float (20)
/Acquisition/SpatialSamplingInterval	Spatial sampling resolution (m)	Float (1.02)
/Acquisition/VendorCode	Manufacturer identifier	String
/Acquisition/Raw[0]/RawData	Raw DAS signals	Integer, (Time × 1663)
/Acquisition/Raw[0]/RawDataTime	Timestamps for raw data	Integer, (Time)
/Acquisition/Raw[0]/Custom/GpBits	Post-processing status flags	Integer, (Time)
/Acquisition/Raw[0]/Custom/GpsStatus	GPS synchronization status	Integer, (Time)
/Acquisition/Raw[0]/Custom/PpsOffset	Timing offset (clock units)	Integer, (Time)
/Acquisition/Raw[0]/Custom/SampleCount	Number of collected samples	Integer, (Time)

Table 2. Overview of HDF5 Datasets and Attributes.

Metric	Value
Accuracy	91.4 %
Macro-averaged Precision	85.7 %
Macro-averaged Recall	88.2 %
Macro-averaged F1-Score	86.8 %

Table 3. Test metrics of the CNN model.

Argument	Description
data_dir	Path to the main dataset directory containing subdirectories. Each subdirectory name corresponds to a label name.
sample_len	Length of the sample, which must be smaller than the window size. It is used to trim the sample after applying the transform.
transform	A preprocessing method applied to each signal window from the dataset.
fsize	Size of the window in samples.
shift	Overlap between windows in samples.
drop_noise	Flag to indicate whether to drop noisy samples.
decimate	Dictionary specifying the decimation factor for each label.

Table 4. Main arguments of the DASDataLoader class.

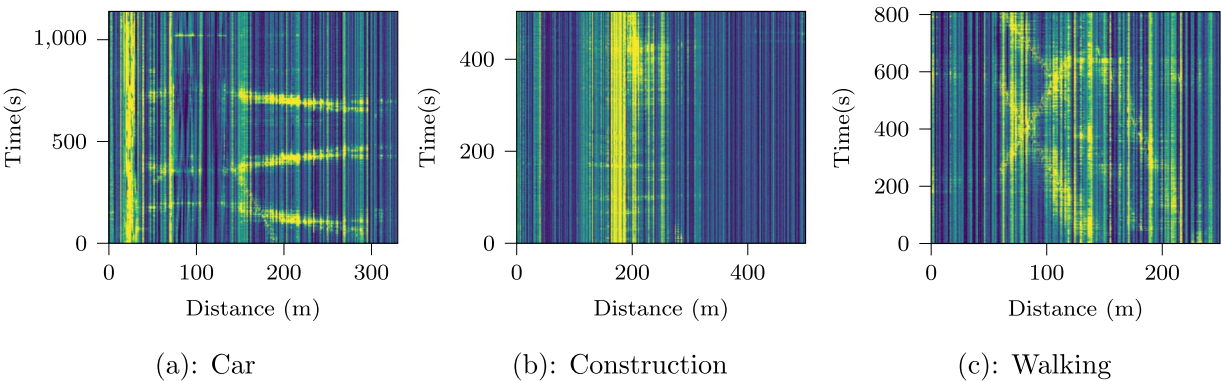


Fig. 3 Band-limited signal power P_{band} representations for different labeled events: car, construction and walking. Each subfigure highlights the distinct acoustic signatures captured in the dataset.

demonstrates its suitability for advanced analytical applications and reinforces its potential utility in related fields.

Beyond its application for event classification, the dataset provides a valuable resource for training deep learning models in feature extraction tasks, particularly in DAS applications. By leveraging CNNs, models can learn distinct vibration patterns associated with different activities, facilitating robust representation learning. Furthermore, as demonstrated in prior work on object classification in urban environments using DAS, transfer learning can be applied to adapt models trained in one environment (e.g., a university campus) to another location (e.g., a city center) with minimal retraining¹¹.

To illustrate the quality of the labeled data, Fig. 3 presents P_{band} representations for three distinct labeled events: car movement, construction activities, and walking. Each subfigure reveals the unique acoustic signatures associated with these events, demonstrating the DAS system's ability to capture and distinguish between them. The P_{band} visualization for car movement shows a concentrated pattern of acoustic energy across specific frequency bands, reflecting the consistent vibrations and noise levels generated by vehicles passing near the sensing fiber. In contrast, the P_{band} for construction activities displays a broader distribution of acoustic energy, indicating the variety of noises produced during such activities, including machinery operation and ground vibrations. The walking event's P_{band} is characterized by intermittent bursts of acoustic energy, corresponding to the periodic impact of footsteps, which is less dense than the other activities, illustrating the subtler nature of walking.

Usage Notes

To facilitate access to the labeled dataset, a Python class named **DASDataLoader** is provided in the `data_loader.py` file. This class offers a straightforward way to load and manipulate the labeled data. The arguments for the class constructor are detailed in Table 4.

Two key arguments for modifying the data are `decimate` and `transform`. Loading all samples into computer memory would require an enormous amount of memory, which can be mitigated by decimating the samples. The `decimate` argument accepts a Python dictionary where the key is the exact label name and the value is the decimation factor. An example of how to use the dataloader object with these arguments is provided in a simple example located in the dataset folder in `python/example.py`.

Code availability

The dataloader is distributed under the GPLv3 license, ensuring that it is free to use, modify, and distribute under the same terms. It is available for download alongside the dataset. This resource is intended to facilitate easy access and integration of the dataset into your research or applications. Should you have any questions or require additional help, please do not hesitate to contact the team via email at.

To effectively utilize the **DASDataLoader** class for managing and processing the labeled and unlabeled datasets, specific Python packages are required. These dependencies are detailed in the Pipfile provided with the module. The Pipfile ensures that all necessary libraries are installed and configured correctly. Below is an overview of the required dependencies:

scikit-learn: This library is crucial for implementing machine learning algorithms and tools. It is used in the data processing and feature extraction stages, particularly for applying transformations and modeling the dataset.

h5py: This package provides an interface to the HDF5 data format, which is employed for reading and writing the dataset files. It facilitates the efficient handling of large-scale data stored in hierarchical structures.

numpy: A fundamental package for numerical computations in Python. It is used extensively in the data manipulation and processing functions within the **DASDataLoader** class.

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Author contributions

P.M., A.T., and T.H. conceived the experiment(s), A.T. and P.D. conducted the experiment(s), and P.Z. and A.T. analyzed the results. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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